SCOTT BERKUN

Menu

The Ten Myths of Innovation: the best summary (Updated)

Posted on March 26, 2013 in Myths of Innovation

Scott Berkun is the author of nine popular books on creativity, design, leadership and public speaking. You can hire him to speak or follow him on LinkedIn and Facebook.

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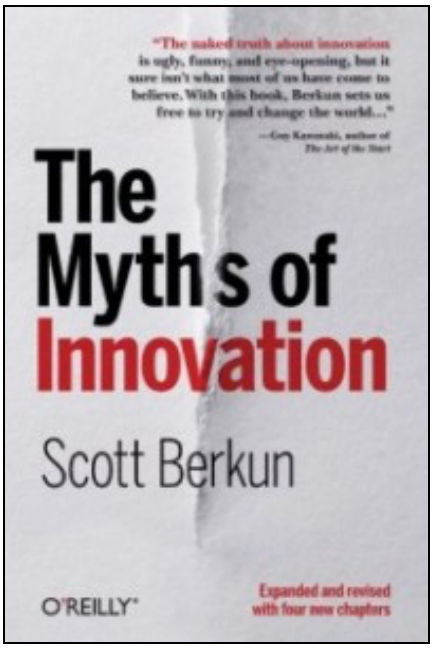
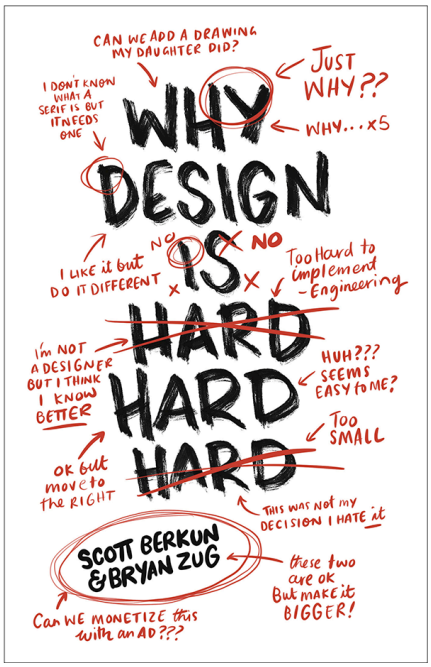


The timeless patterns that explain how innovation happens, or more often doesn't, are explained in the bestselling book I wrote called [The Myths of Innovation](). The way innovation is generally explained is flawed. We rely on legends and myths that sound inspiring but have little relation to the truth about how good ideas become real.

If you're frustrated by how your organization kills good ideas, or relies on caution more than creativity despite the proclamations from leaders to innovate, this book will help you understand what's really going on and why.

The stories and facts in the book are its greatest asset. You'll gain **the longest lasting tools for thinking clearly about innovation from the book itself.**

To help as many people as possible get beyond the myths, they are summarized here. The book was



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heavily researched with 100s of footnotes and references, but here's the tightest summation:

1. **We overvalue the role of flashes of insight (The myth of epiphany)**. Flashes of insight feel great, but the way they're reported distorts the well-documented history of how creativity and success happen. Epiphanies are a consequence of effort, not just the inspiration for it. And no idea is completely original, as all ideas are made from other ideas. When you hear a story about a flash of insight, the useful questions to ask are **1)** how much time the creator spent working before the flash happened, **2)** How many ideas from other people they reused and **3)** how much work they did after the flash to make the idea successful. An epiphany doesn't find investors, make prototypes, sacrifice free time or persist in the face of rejection: only you can do that and you'll have to do it without a guarantee of success (More about epiphany myths).
2. **Technological progress does not move in a straight line (The myth that we know history).** We romanticize the past to fit the present, creating traps for creatives who

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don't know the true history of their own field. Edison did not invent the lightbulb. Ford did not invent the assembly line. Inspiring lies are often more popular than complex truths. And history is heavily tainted by survivorship bias, which distracts us away from more useful historic lessons. History is not a straight line of progress: it wasn't clear that B would follow A, until after they happened, which means the present isn't a straight line of progress either (jetpacks and flying cars are not flukes, historically we're wrong far more often than we think). (see **Myth #8: the best idea wins**).

3. **Progress, and market success, are inherently unpredictable (The myth of a method).** The challenge with creative work, especially in a marketplace, is the many factors beyond your control. You can do everything right and still fail. Most books on creativity make big promises based on history: they cherry pick examples from the past to support their "method". Methods can be useful but they deny that the present is different from the past. There are too many variables in the present to have

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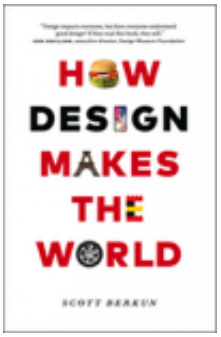

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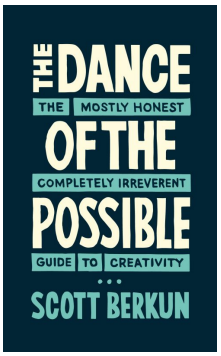

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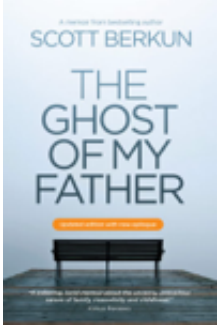
"You'll find a lot to steal from this short, inspiring guide to being creative. Made me want to get up and make stuff!" - Austin Kleon, author of How To Steal Like An Artist

certainty. This is why terms like **innovation system** or **innovation pipeline** are absurd. The idea of an **innovation portfolio,** where a range of risk is assumed across multiple ideas, is more honest. Many books on creativity are surprisingly uncreative (lightbulbs should be banned from creativity book covers) and make impossible promises.

4. **People resist change, including progress (The myth we love new ideas).** We are a conservative species: try something as simple as standing, rather than sitting, in your next group meeting. How accepting were your peers? Conformity is deep in our biology. While talking about creativity is very popular, actually being creative puts your social status at risk. All great ideas were rejected, often for years or decades, yet we bury this in our history (see Myth #1 & #2). The history of breakthroughs is a tale of persistence against rejection. Much of what makes a successful innovator is their ability to persuade and convince conservative people of the merits of their ideas, a very different skill from creativity itself. Your problem is likely not your ideas, but your



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skills for pitching ideas to others. Ideas are rarely rejected on their merits; they're rejected because of how they make people feel. The bigger the idea, the harder the persuasion challenge.

5. **We overstate individual contributions and under-recognize teams (The myth of the lone inventor).** It's easier to worship a hero if they are portrayed as superhuman. But even people worthy of the title genius or prodigy like Mozart, Picasso and Einstein had family and teachers who taught them. Many of Edison's patents are shared with co-workers, as despite his huge ego he knew collaboration was critical (His Menlo Park office was one of the first research labs). Stories of mad geniuses who worked completely alone are rare. Pick any master who you think worked alone and read some of their history: you'll be surprised how many people influenced their work. Learning to collaborate, and give and receive feedback, may matter more than your brilliance.

6. **Good ideas are everywhere, it's courage that's scarce (The myth that good**

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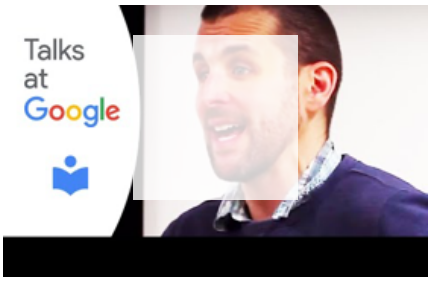
ideas are rare). If you watch any 6 year old child they will invent dozens of things in an hour. We are built for creativity. The problem is the conventions of adult life demand conformity and we sacrifice our creative instincts in favor of social status. Unlike a child, adults are supremely and instantly judgmental, killing ideas before they've had even a moment to prove their worth. It's easy to rediscover creativity, which is why brainstorming rarely helps much. We're already creative. The challenge is **ideas don't come with the courage to invest in them**. Good ideas are everywhere: what's uncommon is people with the conviction to put their reputation behind ideas.



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7. **People in charge resist progress since change threatens the status quo that benefits them (The myth your boss knows more than you).** A fallacy of workplaces is that senior staff are better at everything than the people who work for them. This is false in many ways, but creative intuition might be the most false. To rise in power demands good political judgement, yet innovation requires a willingness to defy

convention. Convention-defiers are harder to promote in most organizations, yet essential for progress. To assume senior staff are the best at leading change is a mistake.

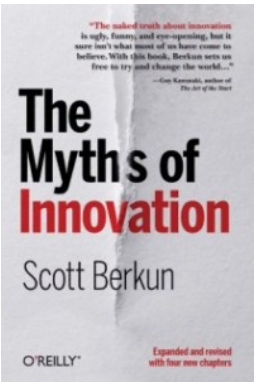
8. **The world of ideas is not a meritocracy (The myth the best idea wins).** We lionize winners and history blames losers for their fate, even if they did most of the same things the winners did (See survivorship bias). Marketing, politics and timing have tremendous influence on why one idea or its competitors wins, yet these details are more complex than we want to hear and fade from history. It's satisfying to believe the best idea has won in the past, because it's something we want to believe about the present too. But to be successful with ideas demands studying why some lousy ideas have triumphed (Why doesn't the U.S. use the metric system?), and some great ones are still on the sidelines. The world of ideas is not a pure meritocracy and you need to act accordingly (See related chapter excerpt).

9. **Defining problems well is as important as solving them (The myth that problems are**

less interesting than solutions). Einstein said "If I had 20 days to solve a problem I would take 19 to define it." There are many creative ways to think about a problem, and different ways to look at a situation. The impatient run at full speed into solving things, speeding right past the insights needed to find a great solution. If you listen to how successful creators talk about their daily work, they spend more time thinking about the problem than the epiphany obsessed media would have us believe.

10. **Unintended consequences are hard to avoid (The myth that innovation is always good).** How would you feel about an invention that ends your profession? What impact will an idea have 1,5,10,100 years from now? All innovation is change and all change helps some people and hurts others. Many horrible inventions were created with the best intentions (and some horrible intentions led to some good consequences). Benz and Ford never imagined automobiles would kill 40k people annually in the U.S. And the Wright brothers never imagined Predator drones. Any successful idea has a multitude of consequences that are

impossible to predict and difficult to even measure.



Click here to Watch the presentation based on the book (OPTIONAL)

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(https://techratchet.com/2021/08/02/book-summary-the-innovators-delimma-by-clayton-christensen/)
- "The Innovator's DNA" by Clayton Christensen et al (https://techratchet.com/2021/10/01/book-summary-the-innovators-dna-by-clayton-christensen-et-al/)
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If you would like to learn more about innovation in history, read my book From Poverty to Progress: How Humans Invented Progress, and How We Can Keep It Going. (https://www.amazon.com/Poverty-Progress-constraints-agriculture-decentralization-ebook/dp/B08RSRY73Q)

PART 2: BRIEF ANALYSIS OF BERKUN'S BOOK

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Topic of Book

Berkun busts a number a widely held myths about innovation.

This book is an excellent introduction to the field of innovation. In contrast to many books in the **field, it is highly readable.**

KEY TAKEAWAYS

- Innovations rarely come from one great epiphany (an "aha" moment). It comes from spending huge amounts of time understanding a field, identifying its key problems and looking for solutions outside the field.
- Traditional histories of innovation are deceptive because they typically only focus on innovations that succeeded and ignore the vast number of similar innovations that failed to gain market share.
- There is no one process for innovation.
- Big ideas in all fields endure dismissals, mockeries, and persecutions (of them and their creators) on their way to changing the world.
- The love of new ideas is a myth: we prefer ideas only after others have tested them. We confuse truly new ideas with good ideas that have already been proven, which just happen to be new to us.
- Innovators rarely find support within mainstream organizations, which tend to be risk averse Inventors rarely work alone, even though history often gives credit to one individual.
- Nearly simultaneous invention by different people without knowledge of each other is a common phenomenon. It's common because innovations demand prerequisite knowledge. The best ideas don't always win. The factors that spread innovations are largely about ease of adoption.
- Successful innovation often involves more attention to problems than solutions.
- Very few innovators studied innovation or creative thinking. They learned by doing.

Important Quotes from Book

One grand myth is the story of Isaac Newton and the discovery of gravity. As it's often told, Newton was sitting under a tree, an apple fell on his head, and the idea of gravity was born. It's entertaining more than truthful, turning the mystery of ideas into something innocent, obvious, and comfortable.

Newton's apple myth is a story of epiphany or "a sudden manifestation of the essence or meaning of something,"1 and in the mythology of innovation, epiphanies serve an important purpose.

Any seemingly grand idea can be divided into an infinite series of smaller, previously known ideas.

For most, there is no singular magic moment; instead, there are many smaller insights accumulated over time.

The myth of epiphany tempts us to believe that the magic moment is the grand catalyst; however, all evidence points to its more supportive role.

One way to think about epiphany is to imagine working on a jigsaw puzzle. When you put the last piece into place, is there anything special about that last piece or what you were wearing when you put it in? The only reason that last piece is significant is because of the other pieces you'd already put into place.

Any major innovation or insight can be seen in this way. It's simply the final piece of a complex puzzle falling into place. But unlike a puzzle, the universe of ideas can be combined in an infinite number.

Epiphany had three parts, roughly described as early, insight, and after. During the early period, hours or days are spent understanding the problem and immersing oneself in the domain. An innovator might ask questions like "What else in the world is like this?" and "Who has solved a problem similar to mine?", learning everything he can and exploring the world of related ideas. And then there is a period of incubation in which the knowledge is digested, leading to experiments and rough attempts at solutions. Sometimes there are long pauses during incubation when progress stalls and confidence wanes.

Deep quiet periods, time spent doing unrelated things, often helps new ideas surface.

The best lesson from the myths of Newton and Archimedes is to work passionately but to take breaks.

To focus on the magic moments is to miss the point. The goal isn't the magic moment: it's the end result of a useful innovation.

Nearly every major innovation of the 20th century took place without claims of epiphany.

The stories told in schools and books present Gutenberg and other innovators as obvious, logical, and necessary contributors to the world, begging the assumption that if we were alive in their time, we'd see them in the same way they're portrayed in our history books. Those glorified accounts present innovation in a distorted way that is impossible to achieve in the present because the neat arcs of progress, clear sense of purpose, and certainty of success are heavily shaped, if not invented, by hindsight.

This means that every technology, from pacemakers to contact lenses, fluorescent lights to birth control pills, arrived through the same chaos seen in the hot technologies of today. Just because dominant designs developed before we were born, or in fields so far from our own that we're ignorant of their struggles, doesn't mean their arrival was predictable, orderly, or even in our best interest. Yet, the dominant designs, the victors of any innovative pursuit, are the ones that get most of history's positive attention.

Not only do timelines express a false omnipotent view of history, they're superficial, offering an illusion of comprehensiveness… At best, timelines show only one path of the full tree of innovation history.

The myth of methodology, in short form, is the belief that a playbook exists for innovation and, like Mr. K.'s deceptively quaint instructions, it removes risk from the process of finding new ideas.

The best advice I've read on starting creative work comes from John Cage, often considered the most innovative composer of the 20th century,5 who said, "It doesn't matter where you start, as long as you start." He meant that there can be no perfect beginning: it's only after you start—no matter how roughly—that you can evaluate and build on what you've done, shift directions, or start over with the insight and perspective you've gained in the process. Innovation is best compared to exploration, and like Magellan or Captain Cook, you can't find something new if you limit your travels to places others have already found.

The majority of innovations come from dedicated people in a field working hard to solve a well-defined problem… the innovators spent time framing the problem, enumerating possible solutions, and then began experimenting.

Many innovations begin with bright minds following their personal interests. The ambition is to pass time, learn something new, or have fun. At some point, the idea of a practical purpose arises, commitments are made, and the rest is history.

Many innovations are driven by the quest for cash.

Waves of innovation have come from individuals in need of something they couldn't find.

Most innovations involve many factors, and it's daft to isolate one above others.

Steve Jobs, founder of Apple and Pixar, was asked, "How do you systematize innovation?" (a common question among CEOs and the business community). His answer was, "You don't."

I've categorized the eight challenges innovators confront:

- Finding an idea
- Developing a solution
- Sponsorsip and funding
- Reproduction
- Reaching potential customers
- Beating competitors
- Timing
- Keeping the lights on

The good news that arises from all of these challenges is that there are many ways to succeed.

Every great idea in history has the big, red stamp of rejection on its face. It's hard to see today because once ideas gain acceptance, we gloss over the hard paths they took to get there.

Big ideas in all fields endure dismissals, mockeries, and persecutions (of them and their creators) on their way to changing the world.

The love of new ideas is a myth: we prefer ideas only after others have tested them. We confuse truly new ideas with good ideas that have already been proven, which just happen to be new to us.

We reuse ideas and opinions all the time, rarely committing to the truly *new*. But we should be proud; it's smart. Why not recycle good ideas and information? Why not take advantage of the conclusions other people have made to efficiently separate what's good and safe from what's bad and dangerous? Innovation is expensive: no one wants to pay the price for ideas that turn out to be not quite ready for prime time.

There is an evolutionary advantage in this fear of new things. Any ancestor who compulsively jumped off every newly discovered cliff or ate only scary-looking plants died off quickly. We happily let brave souls like Magellan, Galileo, and Neil Armstrong take intellectual and physical risks on our behalf, watching from a safe distance, following behind (or staying away) once we know the results. Innovators are the test pilots of life, taking big chances so we don't have to.

The secret tragedy of innovators is that their desire to improve the world is rarely matched by support from those they hope to help.

This creates an unfortunate paradox: the greater the potential of an idea, the harder it is to find anyone willing to try it.

Most have little interest in having their minds changed… This gap—the difference between how an innovator sees her work from how it's seen by others—is the most frustrating challenge innovators face.

Many innovators give up when they learn ideas—even with dazzling prototypes or plans in hand—are only the beginning. The challenges that follow demand skills of persuasion more than brilliance. As Howard H. Aiken, a famous inventor, said, "Don't worry about people stealing

your ideas. If your ideas are any good, you'll have to ram them down people's throats."

The observation many would-be innovators never make is that most criticisms are superficial. The spoken questions only hint at the real concerns. Responding to superficial comments is a loser's game; persuading demands mapping criticisms to deeper issues.

This is the magic double-secret principle: innovative ideas are rarely rejected on their merits; they're rejected because of how they make people feel.

This challenge of mind is known as the *innovator's dilemma*. The face-off between Western Union and Alexander Graham Bell (dramatized but roughly accurate in my telling) has been played out for centuries, with the captains of one aging innovation protecting their work from the threat of emerging ideas.

Frustration with people in power is a perennial complaint among creative minds.

Innovators rarely find support within mainstream organizations.

This explains the natural bond between breakthrough thinkers and new companies: innovative entrepreneurs not only have the passion for new ideas, they also have the conviction to make sacrifices that scare established companies.

The risks for an individual focusing 100% of his resources on a crazy idea are small: it's one life. But for an organization of 500 or 10,000 people, the risks of betting large on a new idea are high.

Most innovations are not the solid, tangible, independent things we imagine them to be. Each one is made up of threads and relationships that don't separate easily or yield simple answers.

Edison, Ford, and countless innovators are recognized as sole inventors for convenience. The histories we know depart from the truth for the simple reason that it makes them easier to remember.

The problem of simultaneous inventorship: the situation when two or more people claim to have invented something.

It's common because innovations demand prerequisite knowledge… Add the limited number of popular problems in any field, and suddenly the number of people chasing particular challenges isn't so large.

Given the combination of shared factors, odds are reasonable that people in the same field, at the same time, studied in the same universities or learned from the same textbooks.11 They might even have mutual friends, drinking buddies, or dance partners, making the chances for simultaneous invention unexpectedly high: as free as people are to think creatively, there is a wardrobe of existing ideas that they're all shopping from.

What makes simultaneous invention (also known as *multiples*) contentious is that creators often work in isolation from—yet in competition with—their peers, making them prone to fantasies that their creations are unique.

Despite the myths, innovations rarely involve someone working alone, and never in history has an invention been made without reusing ideas from the past.

Ideas need nurturing and are grown, not manufactured, which suggests that idea shortages are self-inflicted. It doesn't take a genius to recognize that ideas will always be easier to find if they're not shot down on sight.

The myth that leads to this idea-destroying behavior is that good ideas will look the part when found.

Linus Pauling, the only winner of two solo Nobel Prize awards in history, had this to say about finding ideas: "The best way to have a good idea is to have lots of ideas."

In a recent survey, innovative people—from inventors to scientists, writers to programmers—were asked what techniques they used. Over 70% believed they got their best ideas by exploring areas they were not experts in.

Talent is only as good as the environment it's in… Few managers recognize that their training and experience, designed to protect what exists, work against the forces needed for innovation.

High experience and confidence make people the greatest resisters to new ideas as they have the most to lose.

It's natural for people to protect what they know instead of leaping into the unknown, and managers are no exception. Managers might even be worse, as the politics they rely on to survive can make them more entrenched and defensive.

Professional management was born from the desire to optimize and control, not to lead waves of change.

Good managers of innovation recognize that they are in primary control over the environment, and it's up to them to create a place for talented people to do their best work.

One thing a genius can't do that her manager can is provide cover fire. Whether through power, inspiration, or charisma, managers have the singular burden of protecting their teams. Innovations always threaten someone in power, and executives in search of budget cuts frequently target them first. The manager's unique role is to use whatever means necessary to shield innovation while it's too young to defend itself in the open.

The best ideas don't always win, but that doesn't stop people from believing they should. Most innovators were frustrated by how their ideas, clearly superior in their own minds, struggled for acceptance in the world.

The factors that spread innovations are largely about ease of adoption.

This suggests that the most successful innovations are not the most valuable or the best ideas, but the ones that appear on the sweet spot between what's good from the expert's perspective, and what can be easily adopted, given the uncertainties of all the secondary factors combined. The

idealism of goodness and the notion that goodness wins is tempered by the limits and irrationalities of people's willingness to try new things, the culture of the era, and the events of the time. This explains why the first innovators— driven by complete faith in their ideas—are so often beaten in the market, and in public perception, by latecomers willing to compromise.

Successful innovation often involves more attention to problems than solutions.

One way to creatively describe a challenge is to compare it to another kind of challenge that's been solved.

When considering the creators of the great works of the past, it's surprising how few of them studied innovation or creative thinking.

However, what they did do was pick specific problems they were passionate about, and got to work. They focused on those problems, often with little guarantee of reward. My point is that they didn't seem to need much understanding of innovation as an abstract concept, which many people today believe is the place to start. But a strong case can be made that the opposite is true. Many of the great figures didn't care to study; they preferred to *do*. They quickly got to work trying to solve important problems—that in some cases they thought they could profit from—and learned along the way.

Perhaps the greatest myth of all is that you need to be an expert in innovation in order to change the world.

We like to pretend that gap from knowing to doing is small, but it's enormous, and few people are willing to do the work to close that gap. It requires courage, persistence, comfort with risk, and a willingness to do work with no guaranteed external rewards. These qualities are more important than knowledge, degrees, or shelves of books on innovation.

In this age, being seen as an "expert" may have little bearing on the "expert's" ability to do the thing she is supposedly an expert in. This doesn't mean books are useless, but it does mean there's a paradox: the people most visible for being experts on innovation are those who spend more time writing and talking about it than actually doing it.

Related Books

- "Where Good Ideas Come From" by Steven Johnson (https://techratchet.com/2020/05/21/book-summary-where-good-ideas-come-from-by-steven-johnson/)
- "Adapt: Why Success Always Starts with Failure" by Tim Harford (https://techratchet.com/2020/04/08/book-summary-adapt-why-success-always-starts-with-failure-by-tim-harford/)
- "The Innovator's Dilemma" by Clayton Christensen